LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Tyres and tubes, variation of tyre pressure with all up weight, creep and creep marking on wheels.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Tyres and tubes, variation of tyre pressure with all up weight, creep and creep marking on wheels.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Inspection of wheel assembly for creep marks; verification of pressure against weight chart; safety: wear eye protection and use jack stands.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft tyres?
(b) How does all-up weight affect tyre pressure requirements?
(c) What visual indication suggests a wheel has undergone creep?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Tyre and tube construction types • Function of tyre beads and flanges • Tubeless versus tubed systems • Material properties of tyre carcass • Role of tyre pressure in load bearing	L & Ds	PPT, IB	15 min
(b)	• Pressure variation with all-up weight • Interpretation of pressure/weight charts • Consequences of under-inflation • Consequences of over-inflation • Standard pressure adjustment procedures	L & Ds	PPT, PDF	20 min
(c)	• Definition and causes of wheel creep • Identification of creep marks • Creep marking application standards • Inspection criteria for creep limits • Maintenance actions for crept wheels	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why tyre pressure must increase as all-up weight increases.
(b) Describe the difference between a tubeless and a tubed tyre system.
(c) What maintenance action is required if creep marks exceed allowable limits?

12. Home Assignment and Follow Up	Prepare a brief note on the basic design requirements of a pneumatic system to be discussed in the next lesson (LP 07).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min